NAME _____ SCHOOL _____

BWAILA CLUSTER TERM I EXAMINATIONS

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

PHYSICS

Subject Number: M164/I

Thursday, 8th December, 2022

Time Allowed: 2 hours
8:00 – 10:00 am

PAPER I

(100 marks)

Instructions

1. **This paper contains 11 printed pages. Please check.**
2. Fill in your **name** at the top of each page.
3. This paper contains **two** sections **A** and **B**. In Section **A** there are **ten** short answer questions while in Section **B** there are **three** restricted essay questions.
4. Answer **all** questions in the spaces provided.
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question.
7. In the table provided on this page, **tick** against the number of the question you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Turn over

NAME: _____ CLASS: _____

SECTION A (70 marks)

1. a. (i). State the kinetic theory of matter.

_____

_____

(1 mark)

(ii). Using the kinetic theory of matter, explain how candle wax melts

_____

_____

_____

(2 marks)

b. (i). What is **absolute zero**?

_____

_____

(2 marks)

(ii). Convert **200 K** to degrees Celsius.

(3 marks)

2. a. State Boyle's law

_____

_____

_____

(2 marks)

b. A gas container contains 240 cm^3 of oxygen gas at a temperature of 400 K. What is the temperature of the gas, in degrees Celsius, if the volume is increased to 600 cm^3 at constant pressure?

NAME: _____ CLASS: _____

(3 marks)

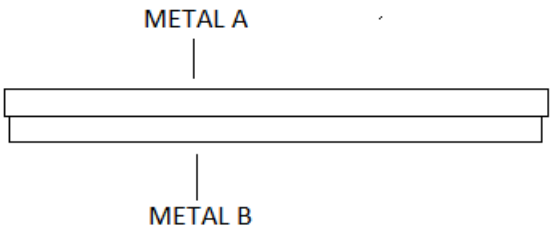
c. Explain, briefly, why tractors have tires with large surface area

_____

_____

(2 marks)

d. The diagram below shows a bimetallic strip. Study it and use to answer the questions that follow. Metal **A** has higher rate of expansion than metal **B**.



The strip is then cooled down to **0C^0**. Draw the strip to show its **new shape** after being cooled.

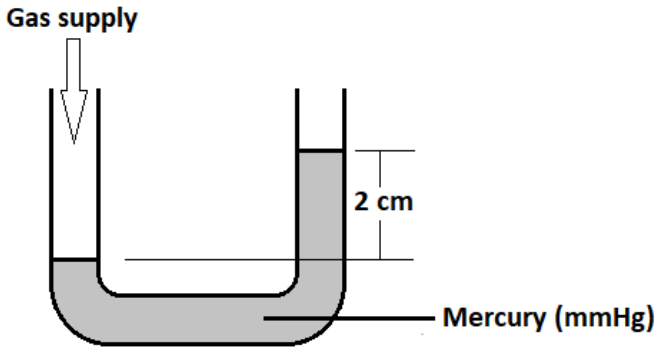
(2 marks)

3. a. A pressure of 49000 Pa is exerted by oil to a bottom of a container. Calculate the depth of the oil in the container if density of the oil is 700 kg/m^3.

(3 marks)

b. The diagram below shows a U-tube manometer. Study it and use to answer the questions that follow.

NAME: _____ CLASS: _____



Calculate the pressure of the gas supply if atmospheric pressure is 740 mmHg

(4 marks)

4. a. Differentiate between distance and displacement

_____

_____

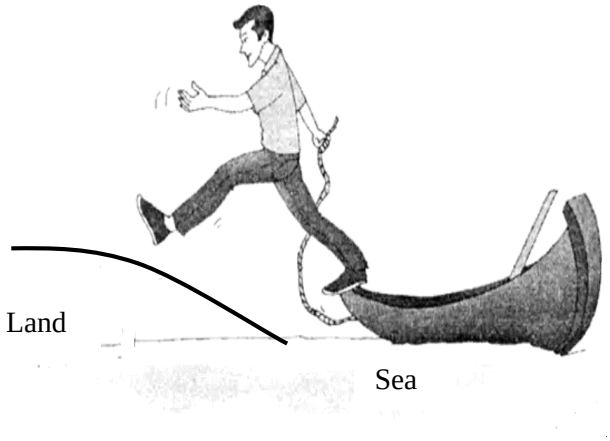
(2 marks)

b. Two forces of 40 N and 60 N are acting at an angle of 60° to each other. Find their resultant force. (Scale: 1 cm to represent 10N)

NAME: _____ CLASS: _____

(4 marks)

c. Study the diagram below and use to answer the questions that follow.



Which **law of motion** is applied in this scenario?

_____

(1 mark)

5. a. State the law of conservation of momentum

_____

_____

(2 marks)

b. A toy car of mass 6 kg moving at 12 m/s collides with a stationary truck of mass 20 kg. If they move together, find their common velocity.

(4 marks)

NAME: _____ CLASS: _____

c. Identify the type of collision in question **5 b.** above.

_____

(1 mark)

6. a. Define the term specific heat capacity.

_____

_____

(2 marks)

b. A block of iron of mass 200 g and specific heat capacity 350 J/kg°C is heated until its temperature changes from 25°C to 70°C. Calculate the amount of heat transferred to the block

(4 marks)

c. Give any **two** applications of specific heat capacity

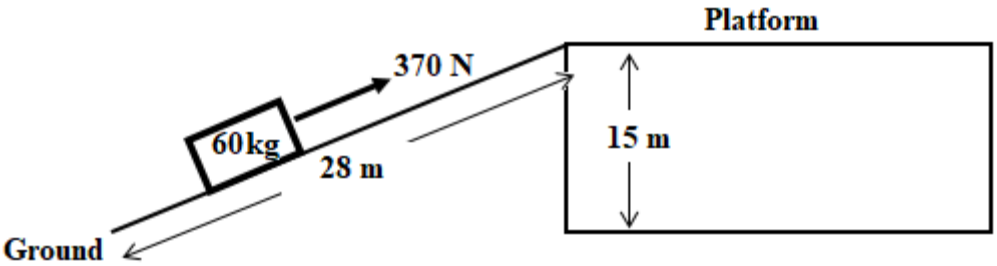
_____

_____

(2 marks)

7. The figure below shows a box being pushed up the inclined plane. Study it to answer the questions that follow.

NAME: _____ CLASS: _____



Calculate the

a. Velocity ratio of the inclined plane

(3 marks)

b. Efficiency of the inclined plane.

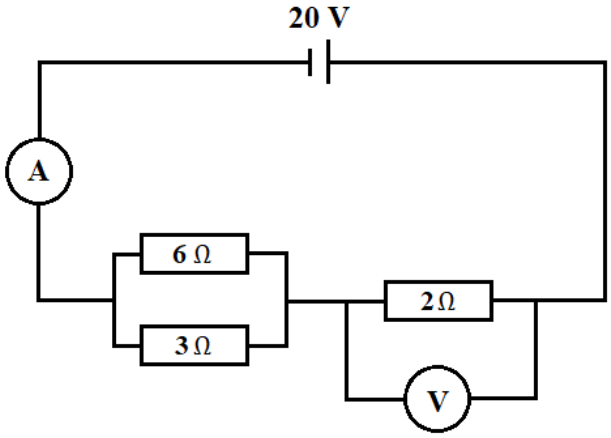
(5 marks)

c. Amount of energy wasted due to friction

NAME: _____ CLASS: _____

(3 marks)

8. The figure below shows a circuit. Study it and use to answer the questions that follow.



a. Calculate the total resistance in the circuit.

(3 marks)

b. Calculate the reading of the ammeter

(2 marks)

9. a. Describe diffraction of a wave.

_____

_____

NAME: _____ CLASS: _____

_____

_____

(3 marks)

b. A turning fork produces 600 waves in 6 minutes. If the speed of the wave is 3.0 m/s, calculate the wavelength of the wave.

(4 marks)

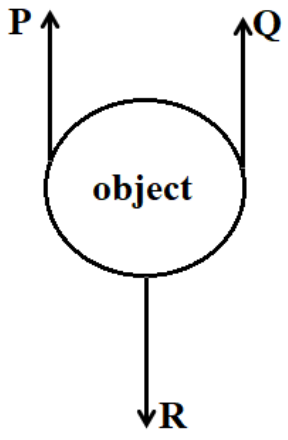
10. a. Define a free fall

_____

_____

(1 mark)

c. Identify the three forces shown on the diagram below



P _____

Q _____

R _____

NAME: _____ CLASS: _____

(3 marks)

SECTION B

(30 marks)

11. a. With the aid of a well labeled diagram sketch a graph and explain how a parachutist falls after jumping from an airplane.

_____

_____

_____

_____

_____

_____

_____

_____

(5 marks)

b. Describe an experiment that can be carried out to find the resistance of a piece of nichrome wire

_____

_____

_____

_____

_____

NAME: _____ CLASS: _____

_____

_____

_____

(5 marks)

12. a. Explain how electric thermostat works to maintain steady temperature in an electric iron

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

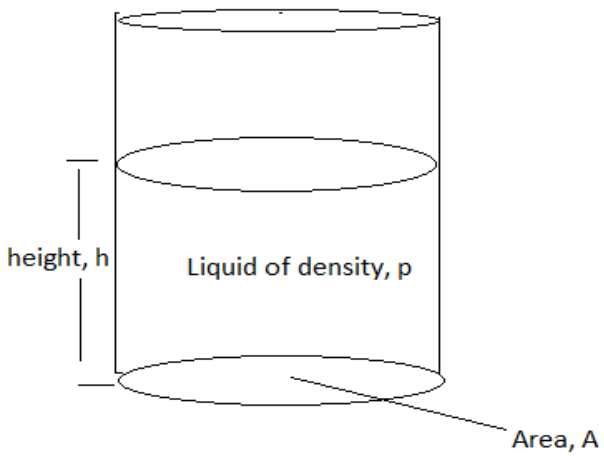
_____

_____

(4 marks)

b. The diagram below shows a tank filled with a liquid of density p filled to a depth of h.

NAME: _____ CLASS: _____



Show that the pressure, $P = \frac{F}{A}$ exerted by the liquid at the base of the tank can be expressed as $\boldsymbol{P = dhg}$

(6 marks)

13. a. With an aid of a diagram, describe how you would demonstrate that a material medium is necessary for the transmission of sound.

NAME: _____ CLASS: _____

_____
_____
_____
_____
_____
_____
_____
_____

(6 marks)

b. Explain how terminal speed is reached by an object that is falling through a liquid.

_____
_____
_____
_____
_____
_____
_____
_____

(4 marks)

END OF QUESTION PAPER